

When Chain-of-Thought Supervision Hurts Scientific-Writing Judges

Anonymous ACL submission

Abstract

Working draft. This manuscript investigates when chain-of-thought supervision helps or harms language-model judges for scientific-writing evaluation. The final abstract will be written after the main results and mechanism analyses are frozen.

1 Introduction

Large language models (LLMs) are increasingly used to score scientific text against task-specific criteria. A common design asks the judge to generate a rationale before returning its score, with the expectation that explicit reasoning will support a more reliable judgment. Yet the available evidence does not show a uniform scoring benefit. Explanation requests can change agreement with human ratings, and chain-of-thought (CoT) can even degrade the judgment distribution of an LLM judge (Chiang and Lee, 2023b; Wang et al., 2025). Likewise, rationale-augmented fine-tuning has improved some tasks but trailed label-only training on others (Hsieh et al., 2023; Shi et al., 2025a; Su et al., 2026). Thus, producing a plausible rationale does not by itself establish that the associated score is learned more reliably.

This inconsistency motivates a more precise question: when a judge outputs both a rationale and a score, which part of the pipeline accounts for any change in scoring? Rationale-first inference asks a fixed model to reason before answering, whereas rationale supervision also optimizes the rationale tokens during fine-tuning. Conflating the two can obscure whether a score changes because the model learned a better criterion-specific decision, or because it became adapted to an inference interface that expects reasoning. We therefore treat rationale supervision and rationale-first inference as separate experimental factors rather than as a single CoT intervention.

Existing generative judges do not provide this separation. Prometheus learns rubric-conditioned feedback followed by a score, and SciRM trains reasoning reward models for multiple scientific-writing criteria (Kim et al., 2024a; Şahinuç et al., 2026). More closely, work on natural language understanding compares label-only and rationale-augmented fine-tuning and proposes Align to balance paired direct and reasoning views (Shi et al., 2025a). However, its evidence comes from general understanding tasks rather than rubric-conditioned scientific-writing evaluation. Whether the same supervision choices improve scientific-writing scores, and how they interact with the inference format, therefore remains unresolved.

In this work, we conduct a controlled study of whether CoT supervision improves scientific-writing judgments under explicit rubrics. Concretely, we train matched judges with label-only targets, CoT targets, or Align, which pairs direct and reasoning supervision. Rather than treating CoT as a single design choice, we independently vary the training target and the inference interface. We apply this design to seven tasks spanning related-work evaluation and review utility, each with an explicit criterion and label space. This separation allows us to identify whether a scoring change reflects supervision, reliance on a reasoning-first interface, or their interaction. We further trace paired prediction changes to distinguish invalid output schemas from shifts among valid scores.

Our contributions and findings are threefold. First, we systematically compare label-only and CoT supervision across multiple scientific-writing criteria and show that CoT is not a uniform improvement: its effects vary by task and inference interface. Second, we demonstrate that performance degradation has two distinct forms, output-schema failure and valid-score drift, which are obscured by aggregate metrics. Third, we adapt Align to preserve direct supervision while learning a rea-

soning path. Align recovers roughly three quarters of the label-only-correct predictions lost after CoT fine-tuning across two seeds, although our current evidence does not attribute this recovery solely to its loss-balancing design. Together, these findings show that rationale generation and scoring reliability must be evaluated as related but distinct objectives.

2 Related Work

2.1 Evaluating Scientific Writing

Scientific-writing evaluation extends beyond surface fluency to evidence synthesis, contribution positioning, and useful criticism. Prior work suggests that LLM-generated feedback can overlap with human reviews and help researchers, while focused-feedback systems target specific, actionable, and coherent comments (Liang et al., 2024; Chamoun et al., 2024). However, evidence about generated feedback does not directly establish the reliability of rubric-conditioned scores.

Recent work evaluates scientific writing through task-specific, verifiable judgments. For text revision, conventional similarity metrics and LLM-based judges capture instruction following and correctness unevenly (Jourdan et al., 2025); other benchmarks test limitation identification (Xu et al., 2025), critique grounding (Ou et al., 2025), and alignment of review focus with experts (Shin et al., 2025). Dedicated frameworks further decompose related-work quality into coherence and research positioning (Şahinuç et al., 2025), and review utility into actionability, grounding specificity, helpfulness, and verifiability (Sadallah et al., 2025). SciRM combines these tasks in a multi-task reward model (Şahinuç et al., 2026). Together, these studies show that scientific-writing evaluation is multidimensional rather than reducible to surface similarity or a single score. However, they focus mainly on defining criteria and improving evaluator performance, leaving how supervision and output design shape scoring reliability underexplored.

2.2 Training LLM-as-a-Judge

LLM-as-a-Judge uses LLMs either to score individual outputs directly or to compare output pairs (Chiang and Lee, 2023a; Zheng et al., 2023). G-Eval further generates task-specific evaluation steps before direct scoring (Liu et al., 2023). However, explanation requests and candidate order can alter agreement with human ratings and pairwise prefer-

ences (Chiang and Lee, 2023b; Wang et al., 2024).

Fine-tuned open judges provide specialized alternatives to closed API judges. Prometheus generates rubric-conditioned feedback and scores, Prometheus 2 supports direct and pairwise judgments, and FLAMe scales training across more than 100 evaluation tasks (Kim et al., 2024a,b; Vu et al., 2024). TRACT further augments joint CoT-score cross-entropy with a regression-aware score loss (Chiang et al., 2025). Because these systems vary supervision, objectives, and inference formats together, they do not isolate whether rationale supervision improves binary or ordinal scoring.

2.3 Rationales in Training and Inference

Rationale-first inference and rationale supervision intervene at different stages. The former prompts a fixed model to generate reasoning before its judgment, whereas the latter optimizes rationale tokens during fine-tuning. Prompted CoT yields its clearest gains on mathematical and symbolic tasks (Sprague et al., 2025); in LLM-as-a-Judge, it can instead narrow the judgment distribution and degrade performance (Wang et al., 2025). Thus, rationale-first inference does not uniformly improve scoring, nor does any inference-time gain establish that rationales are beneficial training targets.

Training-time evidence is similarly conditional. Distilling Step-by-Step treats label prediction and rationale generation as separate multi-task objectives and improves data efficiency (Hsieh et al., 2023). In contrast, naive rationale-augmented SFT can trail label-only training on NLU tasks and harm clinical prediction, whereas objectives that protect answer-token learning can recover accuracy (Shi et al., 2025a; Su et al., 2026; Shi et al., 2025b). Together with generative judges such as TRACT (Chiang et al., 2025), these results leave unclear whether scoring differences arise from rationale supervision, the score objective, or compatibility between training and inference formats. We separate these factors by crossing score-only and rationale-supervised fine-tuning with direct and rationale-first inference while holding the data, labels, and evaluation tasks fixed.

3 Experiment Setup

3.1 Models

We use Qwen3-4B (Yang et al., 2025) as the primary backbone and fine-tune it separately on each scientific-writing evaluation task. For comparison,

we evaluate SciRM (Şahinuç et al., 2026), a reward model developed specifically for scientific-writing assessment, as well as several third-party large language models under the same evaluation protocol.

3.2 Dataset

Following Şahinuç et al. (2026), we construct the evaluation suite from two complementary sources. The related-work tasks are based on the fine-grained evaluation framework introduced by Şahinuç et al. (2025) and extended by Şahinuç et al. (2026) to cover additional machine-learning venues. The review-utility tasks are drawn from RevUtil (Sadallah et al., 2025) and focus on the quality of peer-review comments.

Related-work data

The related-work data evaluate literature synthesis along three dimensions: coherence, positioning type, and positioning check. Each task is formulated as a binary classification problem, with score 0 indicating that the specified criterion is not satisfied and score 1 indicating that it is satisfied.

- (1) *coherence* (Şahinuç et al., 2025): Determines whether a citation sentence accurately reflects and remains consistent with the findings of the cited paper.
- (2) *positioning type* (Şahinuç et al., 2025): Determines whether the related-work section presents the paper’s contribution or positioning in the required location and form.
- (3) *positioning check* (Şahinuç et al., 2025): Determines whether the related-work section compares the target paper with prior studies and establishes its position in the literature.

Review-utility data

The review-utility data evaluate peer-review comments along four dimensions: actionability, grounding specificity, helpfulness, and verifiability. These tasks use five-point ordinal scales, with scores from 1 to 5 and higher scores indicating more useful, specific, actionable, and better-supported feedback.

- (1) *actionability* (Sadallah et al., 2025): Measures how explicitly and concretely a review comment specifies the changes that authors should make.

- (2) *grounding specificity* (Sadallah et al., 2025): Measures whether a review comment identifies the relevant part of the paper and specifies the issue that should be addressed.
- (3) *helpfulness* (Sadallah et al., 2025): Measures whether a review comment provides clear, constructive, and actionable guidance for improving the paper.
- (4) *verifiability* (Sadallah et al., 2025): Measures whether the claims in a review comment are supported by sufficient reasoning, evidence, or external references.

The processing of the source data and the construction of the final training and test sets are detailed in Appendix A. Representative examples from the resulting datasets are provided in Appendix B.

4 Main Results

To be continued...

5 Behavioral Mechanism Analysis

To be continued...

6 Discussion

To be continued...

7 Conclusion

To be continued...

Limitations

To be continued...

Ethical Considerations

To be continued...

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A Dataset Process

A.1 Source Data and Data Cleaning

We obtained the original data from Şahinuç et al. (2026) and retained three related-work and four review-utility dimensions, excluding *verifiability extraction*. We removed duplicate, overlapping, malformed, and unreliable instances. RevUtil is substantially larger than the related-work data; we therefore reproducibly sampled 4,800 candidates per RevUtil dimension while approximately preserving the original label distributions.

A.2 Training and Test Data Construction

Zero-shot prompt construction. The original prompts included fixed few-shot examples containing both rationales and labels, which could influence the model’s reasoning and confound comparisons across supervision settings. As illustrated in Figure 1 and Figure 2, we removed the example block and its corresponding system instruction while leaving the query, criteria, scoring rubric, and evaluation instance unchanged.

Teacher rationale distillation. We use DeepSeek-V4-Pro to generate a rationale and score for each training candidate under the configuration summarized in Table 1. A trajectory is retained only if it strictly follows the required output format, contains a score from the task-specific label space, and assigns the same score as the gold label.

Configuration	Value
Teacher model	DeepSeek-V4-Pro
Temperature	0
Top- <i>p</i>	1
Maximum output tokens	2,048
Number of completions	1
Output format	<reasoning>...</reasoning> followed by <score>N</score>
Acceptance criterion	Format valid and score matches gold label

Table 1: Configuration used for teacher-rationale distillation.

CoT and label-only data construction. The final training data contain two supervision formats. CoT data pair each evaluation prompt with a teacher-generated rationale followed by the gold score, whereas label-only data pair the prompt with the gold score alone. Figures 3 and 4 present the two formats using the same RevUtil actionability instance.

A.3 Final Dataset Statistics

Table 2 reports the task format and final numbers of training and test instances for all seven datasets, together with their aggregate sizes.

Task	Format	Train	Test
Related Work–Coherence	Binary	1,526	1,046
Related Work–Positioning Check	Binary	2,666	603
Related Work–Positioning Type	Binary	944	204
RevUtil–Actionability	Ordinal	1,788	1,000
RevUtil–Grounding Specificity	Ordinal	2,652	1,000
RevUtil–Helpfulness	Ordinal	2,279	1,000
RevUtil–Verifiability	Ordinal	1,825	788
Total	–	13,680	5,641

Table 2: Task formats and final training and test set sizes. Binary tasks use scores of 0–1, whereas ordinal tasks use scores of 1–5.

B Paired CoT and Label-Only Examples

For every evaluation type, we construct matched CoT and label-only views from the same underlying instance. The two views preserve the task query, criteria, scoring rubric, instance identifier, and gold label; they differ only in the system instruction and required output format. The CoT view requests a free-text rationale followed by the score, whereas the label-only view requests only the score. Table 3 links to both views for all seven evaluation types.

Evaluation type	CoT	Label-only
RW–Coherence	Fig. 5	Fig. 5
RW–Positioning Type	Fig. 6	Fig. 6
RW–Positioning Check	Fig. 7	Fig. 7
RU–Actionability	Fig. 8	Fig. 8
RU–Grounding Specificity	Fig. 9	Fig. 9
RU–Helpfulness	Fig. 10	Fig. 10
RU–Verifiability	Fig. 11	Fig. 11

Table 3: Index of the matched CoT and label-only evaluation prompts for all seven tasks. Both columns for a task refer to the same underlying instance and link to the corresponding format within the paired prompt figure.

Original prompt

System: You are an evaluator of expert-domain scientific writing. You will receive a query–answer pair, evaluation criteria, a scoring rubric, and examples demonstrating how the evaluation should be performed. First output your reasoning between `<reasoning>` and `</reasoning>`, followed by the numeric score between `<score>` and `</score>`.

User: [QUERY] → [CRITERIA] → [EXAMPLES] → [ANSWER]

Figure 1: Original prompt format. The system prompt announces fixed few-shot examples, and the user prompt includes an example block containing rationales and labels.

Normalized prompt

System: You are an evaluator of expert-domain scientific writing. You will receive a query–answer pair, evaluation criteria, and a scoring rubric. First output your reasoning between `<reasoning>` and `</reasoning>`, followed by the numeric score between `<score>` and `</score>`.

User: [QUERY] → [CRITERIA] → [ANSWER]

Figure 2: Normalized prompt format. The fixed few-shot examples and the corresponding system instruction are removed, while all task-specific content is retained.

CoT training example: RevUtil actionability

System prompt. You are an evaluator of expert-domain scientific writing. You will get a query-answer pair along with criteria explaining the specific evaluation aspect and the scoring rubric. You should evaluate whether the answer satisfy the query based on the given criteria. First output your reasoning enclosed between `<reasoning>` and `</reasoning>`. Then, output your score enclosed between `<score>` and `</score>`. Inside `<score>` provide only the numeric score and nothing else.

User prompt.

[QUERY]: Your task is to write a review comment for a scientific paper. The comment should be actionable. Those actions should be clearly identifiable and concrete.

[CRITERIA]: Explicit actions or suggestions are direct or apparent. Authors can directly identify modifications they should apply to their draft. Clarification questions should be treated as explicit statements if they give a direct action. However, implicit actions need to be inferred from the comment. This includes missing parts that need to be added. Authors can deduce what needs to be done after reading the comment. For concrete actions, the authors know exactly what needs to be done and how to apply the action. However, for vague actions the authors still do not know how to carry out this action. Scoring rubric is as follows:

- 1: The comment lacks meaningful information to help authors improve the paper. Authors do not know what they should do after reading the comment.
- 2: The comment includes an implicitly stated action or an action that can be inferred. However, the action itself is vague and lacks detail on how to apply it.
- 3: The comment explicitly states an action but is vague on how to execute it.
- 4: The comment implicitly states an action but concretely states how to implement the inferred action.
- 5: The comment contains an explicit action and concrete details on how to implement it. Authors know exactly how to apply it.

[ANSWER]: - line 114: Why not mention that the operator has to be associative and commutative?

Gold label. 5

Completion. `<reasoning>` The comment explicitly suggests adding a mention that the operator must be associative and commutative, specifically at line 114. This is a clear, direct action with a concrete location. The authors know exactly what to write and where, making it fully actionable and concrete. `</reasoning>` `<score>5</score>`

Figure 3: Complete CoT training example for RevUtil actionability.

Label-only training example: RevUtil actionability

System prompt. You are an evaluator of expert-domain scientific writing. You will get a query-answer pair along with criteria explaining the specific evaluation aspect and the scoring rubric. You should evaluate whether the answer satisfy the query based on the given criteria. Output your score enclosed between <score> and </score>. Inside <score> provide only the numeric score and nothing else. Do not output any reasoning or other text.

User prompt.

[QUERY]: Your task is to write a review comment for a scientific paper. The comment should be actionable. Those actions should be clearly identifiable and concrete.

[CRITERIA]: Explicit actions or suggestions are direct or apparent. Authors can directly identify modifications they should apply to their draft. Clarification questions should be treated as explicit statements if they give a direct action. However, implicit actions need to be inferred from the comment. This includes missing parts that need to be added. Authors can deduce what needs to be done after reading the comment. For concrete actions, the authors know exactly what needs to be done and how to apply the action. However, for vague actions the authors still do not know how to carry out this action. Scoring rubric is as follows:

- 1: The comment lacks meaningful information to help authors improve the paper. Authors do not know what they should do after reading the comment.
- 2: The comment includes an implicitly stated action or an action that can be inferred. However, the action itself is vague and lacks detail on how to apply it.
- 3: The comment explicitly states an action but is vague on how to execute it.
- 4: The comment implicitly states an action but concretely states how to implement the inferred action.
- 5: The comment contains an explicit action and concrete details on how to implement it. Authors know exactly how to apply it.

[ANSWER]: - line 114: Why not mention that the operator has to be associative and commutative?

Gold label. 5

Completion. <score>5</score>

Figure 4: Complete label-only training example for the same RevUtil actionability instance as Figure 3.

Related Work–Coherence Shared system instruction. You are an evaluator of expert-domain scientific writing. You will get a query–answer pair along with criteria explaining the specific evaluation aspect and the scoring rubric. You should evaluate whether the answer satisfy the query based on the given criteria. CoT format. First output your reasoning enclosed between <code><reasoning></code> and <code></reasoning></code>. Then output your score enclosed between <code><score></code> and <code></score></code>. Inside <code><score></code> provide only the numeric score and nothing else. Label-only format. Output your score enclosed between <code><score></code> and <code></score></code>. Inside <code><score></code> provide only the numeric score and nothing else. Do not output any reasoning or other text. User prompt. [QUERY]: Given a context from a scientific paper, your task is to write a coherent citation sentence that cites the given paper with a specific citation number. The citation number will also be provided with the context. [CRITERIA]: The paper context supports (entails) the citation sentence. When a sentence references multiple papers, the given paper need only support the part associated with its citation number. Scoring rubric: 0: The context does not align with the claim associated with the citation number or with the overall meaning of the citation sentence. 1: The context is compatible with the claim associated with the citation number. [CONTEXT]: This paper proposes a 3D shape descriptor network, which is a deep convolutional energy-based model for modeling volumetric shape patterns. Maximum-likelihood training follows an “analysis by synthesis” scheme and can be interpreted as mode seeking and mode shifting. The model can synthesize 3D shapes through MCMC sampling, train a 3D generator through MCMC teaching, and support 3D object recovery and super-resolution. Experiments show that it generates realistic shapes and supports 3D shape analysis. CITATION NUMBER: 5 [ANSWER]: Recent work has introduced the EBM with a ConvNet potential [4], [5].
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Figure 5: Matched CoT and label-only evaluation prompts for Related Work–Coherence. The task input is shared across the two formats.

Related Work–Positioning Type

Shared system instruction. You are an evaluator of expert-domain scientific writing. You will get a query–answer pair along with criteria explaining the specific evaluation aspect and the scoring rubric. You should evaluate whether the answer satisfy the query based on the given criteria.

CoT format. First output your reasoning enclosed between `<reasoning>` and `</reasoning>`. Then output your score enclosed between `<score>` and `</score>`. Inside `<score>` provide only the numeric score and nothing else.

Label-only format. Output your score enclosed between `<score>` and `</score>`. Inside `<score>` provide only the numeric score and nothing else. Do not output any reasoning or other text.

User prompt.

[QUERY]: Your task is to write a related-work section for a scientific paper that states the paper’s contribution or its position among the literature in each paragraph.

[CRITERIA]: In a multi-paragraph related-work section, every paragraph should state the paper’s contribution and/or position in the literature. The contribution or positioning statement should align with the specific focus of its paragraph. Scoring rubric:

0: The answer fails to state the paper’s contribution and/or position in each related-work paragraph.

1: The answer states the paper’s contribution and/or position in each related-work paragraph.

[ANSWER]: In several machine-learning settings, negative samples are produced from a statistical generative model. [1] generates negative data using GANs for semi-supervised learning and novelty detection. [2] proposes a theoretical framework that relies on an oracle classifying a sample as valid or invalid, without providing a practical implementation. [3] uses adversarial training to generate hard negatives for NLP tasks. [4] employs a GAN to learn the negative-data distribution for positive-unlabeled learning.

In contrastive unsupervised learning, negative examples are assumed to be semantically farther from positive samples. Word2Vec [5] treats samples from a different context as negatives, while CPC-based methods [6], such as momentum contrast [7], use augmentations of different images. Negative data augmentation generates negatives by transforming existing samples, thereby incorporating useful inductive biases [8].

Figure 6: Matched CoT and label-only evaluation prompts for Related Work–Positioning Type. The task input is shared across the two formats.

Related Work–Positioning Check

Shared system instruction. You are an evaluator of expert-domain scientific writing. You will get a query–answer pair along with criteria explaining the specific evaluation aspect and the scoring rubric. You should evaluate whether the answer satisfy the query based on the given criteria.

CoT format. First output your reasoning enclosed between `<reasoning>` and `</reasoning>`. Then output your score enclosed between `<score>` and `</score>`. Inside `<score>` provide only the numeric score and nothing else.

Label-only format. Output your score enclosed between `<score>` and `</score>`. Inside `<score>` provide only the numeric score and nothing else. Do not output any reasoning or other text.

User prompt.

[QUERY]: Your task is to write a related-work paragraph for a scientific paper that states the paper’s contribution or its position among the literature.

[CRITERIA]: The generated paragraph should explicitly or implicitly mention the main paper’s contribution or position among existing literature. The contribution and/or positioning statement should align with the specific focus of the paragraph. Scoring rubric:

0: The paragraph fails to explicitly or implicitly mention the main paper’s contribution or position among existing literature.

1: The paragraph explicitly or implicitly mentions the main paper’s contribution or position among existing literature.

[ANSWER]: Another line of research focuses on transforming extracted feature spaces into representations that are more discriminative for novel classes. Approaches that learn such transformations include metric-learning techniques and feature-space adaptation methods [9–12].

Figure 7: Matched CoT and label-only evaluation prompts for Related Work–Positioning Check. The task input is shared across the two formats.

RevUtil–Actionability Shared system instruction. You are an evaluator of expert-domain scientific writing. You will get a query–answer pair along with criteria explaining the specific evaluation aspect and the scoring rubric. You should evaluate whether the answer satisfy the query based on the given criteria. CoT format. First output your reasoning enclosed between <code><reasoning></code> and <code></reasoning></code>. Then output your score enclosed between <code><score></code> and <code></score></code>. Inside <code><score></code> provide only the numeric score and nothing else. Label-only format. Output your score enclosed between <code><score></code> and <code></score></code>. Inside <code><score></code> provide only the numeric score and nothing else. Do not output any reasoning or other text. User prompt. [QUERY]: Your task is to write a review comment for a scientific paper. The comment should be actionable. The actions should be clearly identifiable and concrete. [CRITERIA]: Explicit actions are direct or apparent, whereas implicit actions must be inferred. Concrete actions tell authors exactly what to change and how to implement the change; vague actions do not. Clarification questions count as explicit when they specify a direct action. Scoring rubric: 1: The comment lacks meaningful information that could help the authors improve the paper. 2: The comment implies an action, but the action is vague and lacks implementation details. 3: The comment explicitly states an action but is vague about how to execute it. 4: The comment implies an action and concretely explains how to implement it. 5: The comment explicitly states an action and provides concrete details for implementing it. [ANSWER]: 5. Why is TS-CAN better than MTTS-CAN in some cases?
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Figure 8: Matched CoT and label-only evaluation prompts for RevUtil–Actionability. The task input is shared across the two formats.

RevUtil–Grounding Specificity

Shared system instruction. You are an evaluator of expert-domain scientific writing. You will get a query–answer pair along with criteria explaining the specific evaluation aspect and the scoring rubric. You should evaluate whether the answer satisfy the query based on the given criteria.

CoT format. First output your reasoning enclosed between `<reasoning>` and `</reasoning>`. Then output your score enclosed between `<score>` and `</score>`. Inside `<score>` provide only the numeric score and nothing else.

Label-only format. Output your score enclosed between `<score>` and `</score>`. Inside `<score>` provide only the numeric score and nothing else. Do not output any reasoning or other text.

User prompt.

[QUERY]: Your task is to write a review comment for a scientific paper. The comment should refer to a specific part of the paper and clearly identify the issue with that part.

[CRITERIA]: A fully grounded comment allows the author to pinpoint the section, table, figure, or unique aspect being addressed. A specific comment explains what is wrong or missing in that part and provides concrete examples when external work is mentioned. Scoring rubric:

- 1: The comment is neither grounded nor specific.
- 2: The addressed part cannot be identified confidently, and the issue is not specified.
- 3: The addressed part cannot be identified confidently, but the issue is specified clearly.
- 4: The addressed part is explicit or obvious, but the issue is not specified.
- 5: The addressed part is explicit or obvious, and the issue is specified.

[ANSWER]: 4. Figure 3 lacks a description in the main body.

Figure 9: Matched CoT and label-only evaluation prompts for RevUtil–Grounding Specificity. The task input is shared across the two formats.

RevUtil–Helpfulness Shared system instruction. You are an evaluator of expert-domain scientific writing. You will get a query–answer pair along with criteria explaining the specific evaluation aspect and the scoring rubric. You should evaluate whether the answer satisfy the query based on the given criteria. CoT format. First output your reasoning enclosed between <code><reasoning></code> and <code></reasoning></code>. Then output your score enclosed between <code><score></code> and <code></score></code>. Inside <code><score></code> provide only the numeric score and nothing else. Label-only format. Output your score enclosed between <code><score></code> and <code></score></code>. Inside <code><score></code> provide only the numeric score and nothing else. Do not output any reasoning or other text. User prompt. [QUERY]: Your task is to write a review comment for a scientific paper. The comment should be useful for helping the authors improve the paper. [CRITERIA]: A helpful review should be actionable, grounded in a specific part of the paper, and provide justification or evidence for its claims. Scoring rubric: 1: The comment identifies no meaningful weakness or improvement and provides no actionable feedback. 2: The comment identifies an issue but is vague, unclear, or minimally useful. 3: The comment identifies an issue or improvement but is incomplete or lacks depth. 4: The comment provides clear and actionable feedback but could be more comprehensive. 5: The comment thoroughly identifies weaknesses and provides detailed, actionable, and constructive suggestions. [ANSWER]: 1) Which model is better: HMM/DNN, CTC, or Enc-Dec?

Figure 10: Matched CoT and label-only evaluation prompts for RevUtil–Helpfulness. The task input is shared across the two formats.

RevUtil–Verifiability Shared system instruction. You are an evaluator of expert-domain scientific writing. You will get a query–answer pair along with criteria explaining the specific evaluation aspect and the scoring rubric. You should evaluate whether the answer satisfy the query based on the given criteria. CoT format. First output your reasoning enclosed between <code><reasoning></code> and <code></reasoning></code>. Then output your score enclosed between <code><score></code> and <code></score></code>. Inside <code><score></code> provide only the numeric score and nothing else. Label-only format. Output your score enclosed between <code><score></code> and <code></score></code>. Inside <code><score></code> provide only the numeric score and nothing else. Do not output any reasoning or other text. User prompt. [QUERY]: Your task is to write a review comment for a scientific paper. The claims in the comment should be justified through logical reasoning, common sense, or external sources. [CRITERIA]: A claim may be justified through supporting logic, common knowledge in the field, or relevant external references. Scoring rubric: 1: The comment contains a claim without supporting evidence or justification. 2: The comment provides vague, insufficient, or incomplete support. 3: The comment provides support but omits important examples, explanations, or references. 4: The claim is sufficiently supported but has minor gaps. 5: The claim is thoroughly supported by explicit and robust reasoning, references, data, or common-sense arguments. [ANSWER]: 4. It is probably better to add a conclusion section.

Figure 11: Matched CoT and label-only evaluation prompts for RevUtil–Verifiability. The task input is shared across the two formats.